

NETWORKED CONTROL

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EXERCIZES

1) Consider the following continuous-time system $\mathcal{S}$.

$$\begin{cases} \dot{x}_1(t) = x_1(t) + 2u(t) \\ \dot{x}_2(t) = -x_2(t) + x_1(t) + u(t) \\ y(t) = x_1(t) \end{cases}$$

- A. Compute the eigenvalues and the modes of the system. Is the system open-loop asymptotically stable?
- B. We aim to control the system using a digital quantized controller. Answer the following questions.
 - a. According to the theory, what is the minimum rate N_B/h (where N_B is the number of bits encoding the control law and h is the sampling time) required to make the system boundable?
 - b. In view of the answer given to the previous question, what is the maximum value of h that can be used to make the system boundable in case $N_B = 2$?
 - c. Set $h = \log_e 2$ s and $N_B = 2$. To control the system, we use the control law defined in case we operate at the rate limits (with $u_{\min} = -1$ and $u_{\max} = 1$) applied to the observable state of the system (i.e., corresponding to $y(t)$). Analyze the trajectories of $y(t)$ using the graphical method (possibly starting from different selected initial conditions). According to the results of the graphical method and to the theory, what is/are the corresponding invariant set/s where $y(t)$ is bounded to lie asymptotically?
 - d. Under the control law defined at point c, define the qualitative asymptotic behavior of both states $x_1(t)$ and $x_2(t)$. Adequately justify the answer.
 - e. Is the control law used at point c. the unique one that allows to steer the output to lie in a bounded set? **Justify adequately the response.**

2) Define, in a concise way, the different **components and sources of delay in control networks**.

3) Define **linear matrix inequalities (LMIs)**.

How do LMIs allow to solve stability analysis problems for linear time invariant systems?

How do LMIs allow to solve stability analysis problems in the context of networked control systems (NCS), e.g., in case of time-varying delays?